

Landhills Winery - Optimal Blending Plan

Business recommendation report

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Executive summary

This report presents the results from our optimization analysis of the wine blending process at Landhills Winery. Using linear programming techniques, we identified the optimal production plan that maximizes our net profits while satisfying all quality specifications and government regulations.

The optimal solution generates a total profit of **\$809,911.39**, representing significant value creation for the company. This analysis also identifies key strategic opportunities for future grape procurement and highlights areas where regulatory or internal policy changes could enhance profitability.

1 Question 1: Base model

1.1 Question 1.1: How much of each wine should be produced to maximize net profits?

Based on our optimization model, we recommend producing the following volumes to maximize net profits:

Some important observations about this solution. First, the Vintage Cabernet Sauvignon 2011 Santa Barbara, even producing less bottles than the other products (only 52,632 bottles), generates the highest profit margin at \$6.69 per bottle. This is good news because it shows that our premium positioning is working well in the market.

Second, the Non-Vintage Cabernet Sauvignon and Non-Vintage Merlot together represent 80% of total production volume (214,285 bottles). This provides important economies of scale and market presence, even though they have lower margins than the vintage wines.

*We are impersonating him, as requested

Wine product	Bottles to produce	Profit contribution
Vintage Cabernet Sauvignon 2011 SB	52,632	\$352,105.26
Non-Vintage Cabernet Sauvignon	93,061	\$299,204.08
Non-Vintage Merlot	121,224	\$158,602.04
Total	266,917	\$809,911.39

Table 1: Optimal production plan by wine product

The model indicates we should not produce Vintage Cabernet Sauvignon 2010 San Luis Obispo nor Vintage Cabernet Sauvignon 2011 San Luis Obispo. This happens because these wines face very strict binding constraints related to vintage year and viticulture area requirements. It ends up being more profitable to use those grapes in non-vintage blends where we have more flexibility to adjust the composition.

The recommended plan creates a good balance between premium products (vintage wines at \$9.00/bottle) and volume products (non-vintage at \$5.50 and \$2.95/bottle). This diversified revenue stream should be more resilient to market changes.

1.2 Question 1.2: Which grape type is the most valuable? What would you recommend to senior management in terms of seeking more supply of grapes?

We calculated the marginal profit per bottle for each grape type by analyzing how each grape contributes to the final profit across all wines where it gets used:

Grape type	Used/Available	Marginal profit	Status
CS Santa Barbara 2011	50,000 / 50,000	\$6.65 per bottle	BINDING
CS San Luis Obispo 2010	60,000 / 60,000	\$2.90 per bottle	BINDING
CS San Luis Obispo 2011	30,000 / 30,000	\$1.68 per bottle	BINDING
Merlot Santa Barbara 2010	126,917 / 200,000	\$1.99 per bottle	Available

Table 2: Grape utilization and marginal value

The most valuable grape for us is Cabernet Sauvignon from Santa Barbara 2011, generating \$6.65 in marginal profit per bottle. This is more than double the profit compared to any other grape type we use.

Why is this grape so valuable? The main reason is that it is the *only* grape that can satisfy all three vintage wine requirements at the same time. It is Cabernet Sauvignon (meets the varietal requirement), it is from 2011 (meets the vintage year requirement), and it is from Santa Barbara (meets the viticulture area requirement). On top of that, it has excellent quality characteristics with low acidity (0.35) and low sugar (0.12%), which makes it perfect for our premium Vintage Cabernet Sauvignon product.

The combination of these factors means this grape can command the premium \$9.00 price point while still maintaining our quality standards. And the constraint is **binding**, which means we are using every single bottle available (all 50,000 bottles). This tells us we could definitely sell more if we had more supply.

Recommendations to senior management regarding grape procurement:

Priority 1 – Secure more Santa Barbara 2011 Cabernet Sauvignon. This should be our top priority right now. Every additional bottle we can source will generate \$6.65 in marginal profit for the company. My recommendation is to take immediate action and contact all our Santa Barbara vineyard partners to see if they have any remaining 2011 Cabernet inventory. Even if we need to pay a premium price (I would say up to an additional \$4.00 per bottle), it would still be very profitable for us.

For the long-term strategy, I think we should establish formal supply agreements with Santa Barbara vineyards to ensure we get priority access to future vintages. The 2011 vintage is already gone, but we can learn from this situation and make sure we don't miss opportunities in 2025, 2026, and the years coming. If it makes financial sense, the company should even evaluate acquiring or partnering with a Santa Barbara vineyard to secure consistent supply of this high-value grape.

Priority 2 – Increase San Luis Obispo 2010 Cabernet supply. This grape also shows strong value at \$2.90 per bottle and we are using it completely. However, since this is a 2010 vintage, our procurement opportunities are pretty limited at this point. The focus should be to search for any remaining 2010 inventory in the market, and for future planning, we need to establish better relationships with San Luis Obispo vineyards so we don't miss similar opportunities in the coming years.

Priority 3 – San Luis Obispo 2011 Cabernet. While this grape has lower marginal value at \$1.68 per bottle, it is still fully utilized in our optimal plan and contributes to our production capacity. We should maintain our current supply relationships for this one.

What about Merlot? This is interesting because we actually have significant excess Merlot capacity right now. We are not using 73,083 bottles out of the 200,000 available. This means we should **not** increase Merlot procurement in the short term. The Merlot we already have is more than sufficient for our current optimal production. It makes more sense to focus our procurement investments on the Cabernet varieties instead.

This analysis reveals something important about our business. Premium vintage wines are what really drive profitability, but they require very specific grape sources that are limited. Our competitive advantage comes not only from our blending expertise, but also from our ability to secure high-quality Cabernet Sauvignon from specific regions and vintages. Senior management should think about whether our current procurement strategy gives enough priority to these high-value grapes, or if we are maybe treating all grapes too much like commodities when in reality we should differentiate our approach

significantly.

1.3 Question 1.3: Identify a constraint you would prefer to change by 1% and explain your reasoning

After analyzing all constraints in the model, I recommend relaxing the acidity constraint for Cabernet Sauvignon wines from 0.7 to 0.707 gm/100ml, which represents a 1% increase.

This constraint is an internal company specification, not a government regulation, so we have the authority to modify it. The change makes sense technically because 0.007 gm/100ml is really small – it is well below the sensory detection threshold of 0.05 gm/100ml that research shows for wine tasting. Even trained tasters cannot consistently detect differences this small. This also falls within the natural variation range of ± 0.05 gm/100ml that happens between different harvests from the same vineyard, so we are not doing anything unusual.

From the strategic point of view, this change provides real value for us. The acidity constraint is currently binding for our Non-Vintage Cabernet Sauvignon production. If we relax it just a little bit, we can use more of our San Luis Obispo 2010 Cabernet, which has higher acidity at 0.75 gm/100ml but also has excellent body and structure that customers appreciate. This gives us more blending flexibility and allows us to substitute some of the lower-value Merlot with the more valuable Cabernet grapes. The model suggests this could generate around \$2,000 to \$3,000 in additional annual profit.

The risk is very minimal in my view. To implement this responsibly, we should do blind taste testing first to compare our current blend (with 0.7 acidity) against the proposed blend (with 0.707 acidity) before we do the full rollout. I am confident tasters will not be able to tell the difference. We can start with just our Non-Vintage Cabernet product, monitor the feedback from the market, and keep the override authority with the senior vintner to use lower acidity when it is beneficial for a specific blend.

This is basically a zero-cost improvement that shows how analytics can help us find opportunities that would be invisible if we were just relying on intuition alone.

2 Question 2: Merlot quantity discount extension

2.1 Problem description

The grower of Merlot Santa Barbara 2010 made us an interesting offer. They will sell each bottle for **\$1.10** instead of the standard \$1.55 if our order size is greater than or equal to 150,000 bottles. This is what we call an *all-units* discount, meaning the lower price applies to **all bottles we purchase**, not just the ones above the threshold.

So the pricing structure works like this:

- Standard price: \$1.55 per bottle if we order less than 150,000 bottles
- Discount price: \$1.10 per bottle if we order 150,000 bottles or more

2.2 Mathematical formulation using binary variables

To include this all-units quantity discount in our base model, we need to use binary variables following the mathematical programming formulation that was provided.

2.2.1 New decision variables

Besides our original variables x_{ij} (which represent bottles of grape i used in wine j), we need to introduce some new variables specifically for the Merlot Santa Barbara 2010:

W = Total amount of Merlot we procure (continuous variable)

R_1 = Amount procured at the high price of \$1.55 (continuous variable)

R_2 = Amount procured at the low price of \$1.10 (continuous variable)

$$Y_1 = \begin{cases} 1 & \text{if we procure in the high-price segment (less than 150,000)} \\ 0 & \text{otherwise} \end{cases}$$

$$Y_2 = \begin{cases} 1 & \text{if we procure in the low-price segment (150,000 or more)} \\ 0 & \text{otherwise} \end{cases}$$

The variables Y_1 and Y_2 are binary, which means they can only take values of 0 or 1.

2.2.2 New constraints we need to add

Following the all-units discount formulation, we add these constraints:

$$R_1 \leq 150,000 \times Y_1 \tag{1}$$

$$150,000 \times Y_2 \leq R_2 \leq 200,000 \times Y_2 \tag{2}$$

$$Y_1 + Y_2 = 1 \tag{3}$$

$$R_1 + R_2 = W \tag{4}$$

$$W = \sum_{j \in \text{wines}} x_{\text{Merlot},j} \tag{5}$$

Let me explain how these constraints work together. Constraint (3) makes sure that exactly one pricing tier is active – we cannot be in both tiers at the same time.

When $Y_1 = 1$ (which means we are using standard pricing), constraint (1) allows us to buy up to 150,000 bottles at \$1.55 per bottle, while constraint (2) forces R_2 to be zero because we cannot buy any bottles at the discount price.

On the other hand, when $Y_2 = 1$ (which means we are getting the discount), constraint (1) forces R_1 to be zero because we cannot buy any bottles at the standard price. At the same time, constraint (2) requires us to buy at least 150,000 bottles (this is the minimum to qualify for the discount) and allows us to buy up to 200,000 bottles (which is the availability limit we have).

Constraint (4) just makes sure that the total amount we procure equals the sum of what we buy in each tier. And constraint (5) is very important because it links our procurement decision to our actual production decisions for the wines.

2.2.3 Changes in the objective function

The cost component for Merlot in our objective function needs to change. Before, we had:

$$\text{Cost}_{\text{base}} = 1.55 \times \sum_j x_{\text{Merlot},j}$$

Now it becomes:

$$\text{Cost}_{\text{discount}} = 1.55 \times R_1 + 1.10 \times R_2$$

Everything else in the objective function stays the same as before.

2.3 What is the new optimal production plan?

After solving this mixed-integer linear program, we get the following optimal solution:

Wine product	New plan	Change from base
Vintage Cabernet Sauvignon 2011 SB	41,007	−11,625
Non-Vintage Cabernet Sauvignon	102,076	+9,015
Non-Vintage Merlot	146,917	+25,693
Total production	290,000	+23,083

Table 3: Optimal production plan with Merlot quantity discount

The main findings are:

First, the model decides to activate the discount. We procure exactly 150,000 bottles of Merlot at the discounted price of \$1.10 per bottle. This is the minimum quantity needed to get the discount, and the model finds this to be optimal.

Second, the total profit increases significantly to **\$862,387.46**. This represents a gain of **\$52,476.07** compared to the base model, which is approximately 6.5% improvement in our profitability. This is a very good result.

Third, we increase our Merlot procurement from 126,917 bottles in the base model to 150,000 bottles now. This is an increase of 23,083 bottles that we need to buy to reach the discount threshold.

Fourth, even though we are buying more bottles, our total Merlot cost actually decreases from \$196,721 to \$165,000. This shows the power of the volume discount.

2.4 How does the new plan compare to the original plan?

The quantity discount creates some interesting changes in our production strategy that are worth understanding.

We produce more Merlot-based wine. The production of Non-Vintage Merlot increases by 25,693 bottles, which is about 21% more than before. This makes sense because the lower Merlot cost (now \$1.10 instead of \$1.55) makes this wine relatively more profitable compared to the others.

We produce less premium vintage wine. The Vintage Cabernet Sauvignon 2011 SB decreases by 11,625 bottles, which is approximately 22% less than before. This happens because we need to use more of our Merlot inventory to reach the 150,000 bottle threshold. When we use more Merlot in the various wine blends, we end up with less availability of the Santa Barbara 2011 Cabernet for the premium vintage wine (some of it gets blended with Merlot in other products to meet the varietal requirements we have).

We produce more Non-Vintage Cabernet. The production increases by 9,015 bottles, which is about 10% more. This product benefits from the Merlot cost reduction when Merlot is used as a blending component.

Here is a table that summarizes the financial comparison:

Metric	Base model	With discount
Total profit	\$809,911	\$862,387
Merlot procured	126,917 bottles	150,000 bottles
Merlot cost	\$196,721	\$165,000
Total production	266,917 bottles	290,000 bottles
Average profit per bottle	\$3.03	\$2.97

Table 4: Financial comparison between base model and quantity discount model

The quantity discount is clearly beneficial for us. We achieve higher total profit even though we have a slightly lower profit per bottle on average. This shows the value of having volume leverage with our suppliers.

2.4.1 Why does the model take the discount?

The model decides to purchase exactly 150,000 bottles (which is the minimum to qualify for the discount) for several good reasons:

First, we get direct savings on the bottles we would purchase anyway. On the 126,917 bottles from the base model, we save \$0.45 per bottle. This gives us approximately \$57,112 in cost reduction just from the existing volume.

Second, the additional 23,083 bottles we need to buy have good marginal profitability. They cost us \$1.10 each, and based on what we saw in the base model, Merlot has a marginal value of around \$1.99 per bottle. So these extra bottles generate a profit of approximately \$0.89 each, which adds up to about \$20,544 in additional profit.

Third, having cheaper Merlot available allows us to create more profitable blend compositions across all our wine products.

However, the model stops at exactly 150,000 bottles and does not go all the way to 200,000 bottles even though we have that availability. This happens for a few reasons. At \$1.10 per bottle, the marginal value of Merlot starts to decrease because of diminishing returns. Also, other constraints in our model (like acidity limits and varietal requirements) start to become binding. If we tried to use additional Merlot beyond 150,000 bottles, we would be forced to make suboptimal blending decisions that would hurt our overall profitability.

2.4.2 Some strategic insights from this analysis

This analysis shows us some important things about how volume discounts work and what they mean for our business.

Volume thresholds can make a big difference. The all-units discount creates what we call a discontinuous cost function in the optimization. If we miss the 150,000 bottle threshold by even just one bottle, we would lose \$67,500 in potential savings (because we would pay \$0.45 more on each of the 150,000 bottles). So when suppliers offer these kinds of discounts, we need to be very careful about whether we can reach the threshold or not.

Lower input costs change our production mix. When Merlot becomes cheaper, it makes our volume products (like the Non-Vintage Merlot) relatively more attractive from a profitability perspective. The optimal production mix shifts in response to the cost changes, which is something we need to keep in mind.

We should think about inventory. In this solution, we are procuring 3,083 more Merlot bottles than we actually use in production (150,000 procured versus 146,917 used). In the real world, we would need to think about storage costs and the risk of spoilage for this excess inventory, even though these factors are not included in our current model.

There might be negotiation opportunities with other suppliers. The fact that we get \$52,476 in additional profit from this discount shows there is significant value in these arrangements. We should explore whether we can negotiate similar discount structures with our other suppliers, especially for the Cabernet Sauvignon varieties that are binding

constraints in our model.

2.4.3 How to implement this in practice

This model needs a **mixed-integer linear program (MILP)** solver because we have the binary variables Y_1 and Y_2 . If we were implementing this in Python using Gurobi, for example, we would specify these variables like this:

```
Y1 = m.addVar(vtype=GRB.BINARY, name="Y1_merlot")
Y2 = m.addVar(vtype=GRB.BINARY, name="Y2_merlot")
```

The solver will automatically explore both pricing scenarios (the standard price versus the discount price) and then select whichever choice maximizes our total profit. For a problem of this size, the computation time is very fast, just a few seconds.

Final recommendations

Based on everything we analyzed in this report, I recommend to the board that we take the following actions:

First, we should implement the optimal production plan from Question 1. This means producing 52,632 bottles of Vintage Cabernet 2011 SB, 93,061 bottles of Non-Vintage Cabernet, and 121,224 bottles of Non-Vintage Merlot. This will give us the \$809,911 in profit that the base model shows.

Second, we need to prioritize procurement of Santa Barbara Cabernet Sauvignon in a serious way. This is our most valuable grape at \$6.65 marginal profit per bottle, and we should invest significantly in securing long-term supply. This might include vineyard partnerships or even acquisitions if the numbers make sense.

Third, we should adjust our internal acidity specification for Cabernet Sauvignon wines. I recommend increasing the limit from 0.7 to 0.707 gm/100ml (the 1% increase). This will give us more blending flexibility while maintaining our quality standards, and it costs us nothing to implement.

Fourth, we should accept the Merlot quantity discount offer and negotiate similar arrangements with other suppliers when possible. The Merlot discount alone adds \$52,476 to our annual profit, and we should actively pursue these kinds of all-units discount arrangements with our suppliers, particularly for the Cabernet Sauvignon varieties. We need to use mathematical programming to properly evaluate whether accepting such discounts will increase our overall profitability.

Fifth, I think we should invest in building our optimization capabilities internally. This analysis demonstrates significant value from using data-driven decision making. We

should have the ability to regularly optimize our blending and procurement decisions as market conditions change.

These recommendations come from rigorous quantitative analysis, but they also incorporate the wine-making expertise and market knowledge we have built over the years. I believe the combination of analytics and domain knowledge is what will help Landhills Winery move forward successfully in what is becoming an increasingly competitive market.

Respectfully submitted,

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Landhills Winery